Discussion: Identifiability and Optimization in Diagonal Kalman Delta Networks

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1 Setup

The diagonal Kalman Delta Network (KDN) predicts a diagonal covariance and uses it to form an uncertainty-dependent delta-rule write. For key $\boldsymbol{k}_t$, transition $\boldsymbol{\alpha}_t$, process noise $\boldsymbol{\omega}_t$, and observation noise r_t,

$$\widehat{\boldsymbol{p}}_t = \boldsymbol{\alpha}_t^2 \odot \boldsymbol{p}_{t-1} + \boldsymbol{\omega}_t, \tag{1}$$

$$\boldsymbol{\kappa}_t = \frac{\widehat{\boldsymbol{p}}_t \odot \boldsymbol{k}_t}{r_t + \sum_i \widehat{p}_{t,i} k_{t,i}^2}. \tag{2}$$

The memory update remains a delta update,

$$\boldsymbol{S}_t = \boldsymbol{D}_t \boldsymbol{S}_{t-1} + \boldsymbol{\kappa}_t \left(\boldsymbol{v}_t - (\boldsymbol{D}_t \boldsymbol{S}_{t-1})^\top \boldsymbol{k}_t\right)^\top, \qquad \boldsymbol{D}_t = \mathrm{diag}(\boldsymbol{\alpha}_t). \tag{3}$$

The implementation additionally tracks diagonal information $\boldsymbol{c}_t = 1/\boldsymbol{p}_t$ with the calibrated projected update

$$\boldsymbol{c}_t = \frac{1}{\widehat{\boldsymbol{p}}_t} + \frac{d_k}{r_t} \boldsymbol{k}_t^2. \tag{4}$$

This is a scan-compatible diagonal-information approximation, not an exact diagonal Kalman posterior.

2 The Scale Symmetry

For any constant $\lambda > 0$, consider

$$\boldsymbol{p}_t' = \lambda \boldsymbol{p}_t, \qquad \boldsymbol{\omega}_t' = \lambda \boldsymbol{\omega}_t, \qquad r_t' = \lambda r_t, \qquad \mu' = \frac{\mu}{\lambda}, \tag{5}$$

where $\mu = \boldsymbol{c}_0$ is the initial information and hence $\boldsymbol{p}_0 = 1/\mu$. The predicted covariance scales as

$$\widehat{\boldsymbol{p}}_t' = \boldsymbol{\alpha}_t^2 \odot (\lambda \boldsymbol{p}_{t-1}) + \lambda \boldsymbol{\omega}_t = \lambda \widehat{\boldsymbol{p}}_t. \tag{6}$$

Consequently, the gain is invariant:

$$\boldsymbol{\kappa}_t' = \frac{\lambda \widehat{\boldsymbol{p}}_t \odot \boldsymbol{k}_t}{\lambda r_t + \sum_i \lambda \widehat{p}_{t,i} k_{t,i}^2} = \boldsymbol{\kappa}_t. \tag{7}$$

The information update also transforms consistently as $\boldsymbol{c}'_t = \boldsymbol{c}_t/\lambda$. Therefore $\boldsymbol{\omega}$, r, and μ contain a common covariance-scale degree of freedom that cannot be identified from the memory output. Positive floors weaken the symmetry near their boundaries but do not remove the general conditioning problem.

The transition $\boldsymbol{\alpha}_t$ is different. It affects both covariance prediction and the memory transition $\boldsymbol{D}_t\boldsymbol{S}_{t-1}$, so it is not part of the exact scale symmetry. Nevertheless, $\boldsymbol{\alpha}_t$ and $\boldsymbol{\omega}_t$ are partially confounded in the covariance path because multiple pairs can produce similar $\widehat{\boldsymbol{p}}_t$.

3 What the Noisy-MQAR Results Establish

The original diagonal model obtains 76.2 ± 22.3 percent over the 5×5 initialization and data grid. Its initialization component of variation (19.2 points) is larger than its data component (11.4 points). This is not uniform under-capacity: many cells reach 88–96 percent, while a few fail severely.

The most informative intervention is isotropic gain initialization. Zeroing the output map of the process-noise projection makes $\boldsymbol{\omega}_t$ uniform across channels at initialization, after which anisotropy is learned. At the same learning rate and with the same optimizer, this changes performance from

$$76.2 \pm 22.3 \quad \longrightarrow \quad 83.7 \pm 10.4, \tag{8}$$

and halves the initialization variation from 19.2 to 9.0 points. It also rescues both severe collapse cells. Thus the current evidence identifies random initial gain anisotropy as the primary failure, not the nominal learning rate or scale symmetry.

The scale symmetry remains relevant as a secondary optimization defect. It creates a flat direction, gives equivalent physical models different raw parameter values, and lets coupled quantities evolve under different optimizer geometries. In the current Muon setup, matrix weights use Muon with adjusted learning rates, while biases and the one-dimensional prior parameter use AdamW. A single printed learning rate therefore does not imply uniform motion of the physical uncertainty variables.

The prior μ is unlikely to explain late failures. It only controls the initial information, and its influence should decay rapidly over a long sequence when process noise has a positive floor. Learning it adds a weakly identified parameter with little expected benefit.

4 Ways to Remove or Reduce the Symmetry

4.1 Fix the observation-noise scale

The strongest and simplest gauge choice is

$$r_t \equiv 1, \qquad \mu \equiv 1. \tag{9}$$

The model then learns process uncertainty in units of observation noise:

$$\boldsymbol{\omega}_t = \omega_{\min} + \mathrm{softplus}(f_\omega(\boldsymbol{x}_t)). \tag{10}$$

Because r can no longer be rescaled, the common covariance-scale symmetry is removed. This parameterization is the preferred baseline unless token-dependent observation reliability is shown empirically to matter.

4.2 Learn bounded relative observation reliability

If heteroscedastic observation noise is useful, it should change confidence relative to a fixed reference rather than learn an unrestricted global scale. One option is

$$r_t = \exp(s \tanh f_r(\boldsymbol{x}_t))\,, \tag{11}$$

with fixed s. For $s = \log 4$, $r_t \in [1/4, 4]$. The projection should have no learnable global bias, and the prior can remain fixed at $\mu = 1$. A soft anchoring penalty can further control drift,

$$\mathcal{L}_{\text{scale}} = \gamma\,(\mathbb{E}[\log r_t])^2\,. \tag{12}$$

This reduces rather than mathematically eliminates every possible input-dependent rescaling, but it prevents the practically harmful unbounded common-scale direction.

4.3 Parameterize a stationary uncertainty level

The $\boldsymbol{\alpha}$–$\boldsymbol{\omega}$ interaction can be made more interpretable with

$$\omega_{t,i} = \omega_{\min} + (1 - \alpha_{t,i}^2)\rho_{t,i}, \qquad \rho_{t,i} = \text{softplus}(f_\rho(\boldsymbol{x}_t)). \tag{13}$$

Here α controls retention and ρ is the uncertainty level toward which prediction moves. This is exactly the completed `DiagKLA1` experiment. Coupling alone changes 76.2 ± 22.3 to 76.9 ± 20.9, which is effectively a wash. The completed `DiagKLA1-zi` experiment combines coupling with isotropic gain initialization and obtains 80.1 ± 13.3, below the simpler `DiagKLA-zi` result of 83.7 ± 10.4. Thus this parameterization has already been tested: it neither replaces isotropic initialization nor improves on free additive process noise once that initialization is used.

5 Recommended Ablation

The transition-coupled process-noise comparison is already complete:

Variant	Free additive ω	$(1-\alpha^2)\rho$
Random anisotropic initialization	76.2 ± 22.3	76.9 ± 20.9
Isotropic gain initialization	83.7 ± 10.4	80.1 ± 13.3

The remaining experiment should isolate covariance-scale choices while keeping free additive ω and isotropic gain initialization fixed:

Variant	μ	r_t	ω_t
A	learned	learned positive	free additive
B	fixed at 1	learned positive	free additive
C	fixed at 1	fixed at 1	free additive
D	fixed at 1	bounded relative	free additive

Run at least the same 5×5 initialization–data grid. Report mean, total standard deviation, initialization variation, non-finite runs, and convergence time. During training, log ω, r, effective write strength, gain anisotropy, the minimum channel gain, and gradient norms for the process-noise, observation-noise, prior, and transition parameters.

If C matches or improves A, the production model should remove both r_{proj} and μ_{param}. If D improves C, retain only bounded relative observation reliability. Only after fixing this gauge is a separate uncertainty-parameter learning-rate sweep readily interpretable.

6 Conclusion

The diagonal KDN has a genuine common-scale non-identifiability in $(\boldsymbol{\omega}, r, \mu)$, but the completed noisy-MQAR experiments show that random initial anisotropy is the dominant observed instability. The appropriate design is therefore twofold: start the gain isotropically, and fix a covariance unit—preferably $r = 1$ and $\mu = 1$—so the model learns only uncertainty ratios that affect its predictions.